OPPORTUNITY BRIEF · SALES AND PRESALES

NIST's chemical fingerprint library enables manufacturers to identify unknown substances for compliance.

Manufacturing × Regulatory compliance and reporting × Knowledge graph and semantic modelling · OS148 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
45 is the world moving	**50** can we play, can we win	**MEDIUM** 0 named in the evidence	**€829m** partial evidence · per year	**LATER** research and standards activi...	**L1** 5 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€1.2bn** €147m – €7.6bn	**€829m** €105m – €5.4bn	**€10.0m** €1.3m – €65.1m	partial
Observed: tendered contract value, annualised	**€20.9m** €20.9m – €56.6m	**€20.9m** €20.9m – €56.6m	**€251k** €251k – €679k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Knowledge graph and semantic modelling	8.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_ain2 · E_AI_TTM	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Artificial intelligence by NACE Rev. 2 activity series E_AI_TTM is used as a proxy for Knowledge graph and semantic modelling: Text and language analysis as the nearest series. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

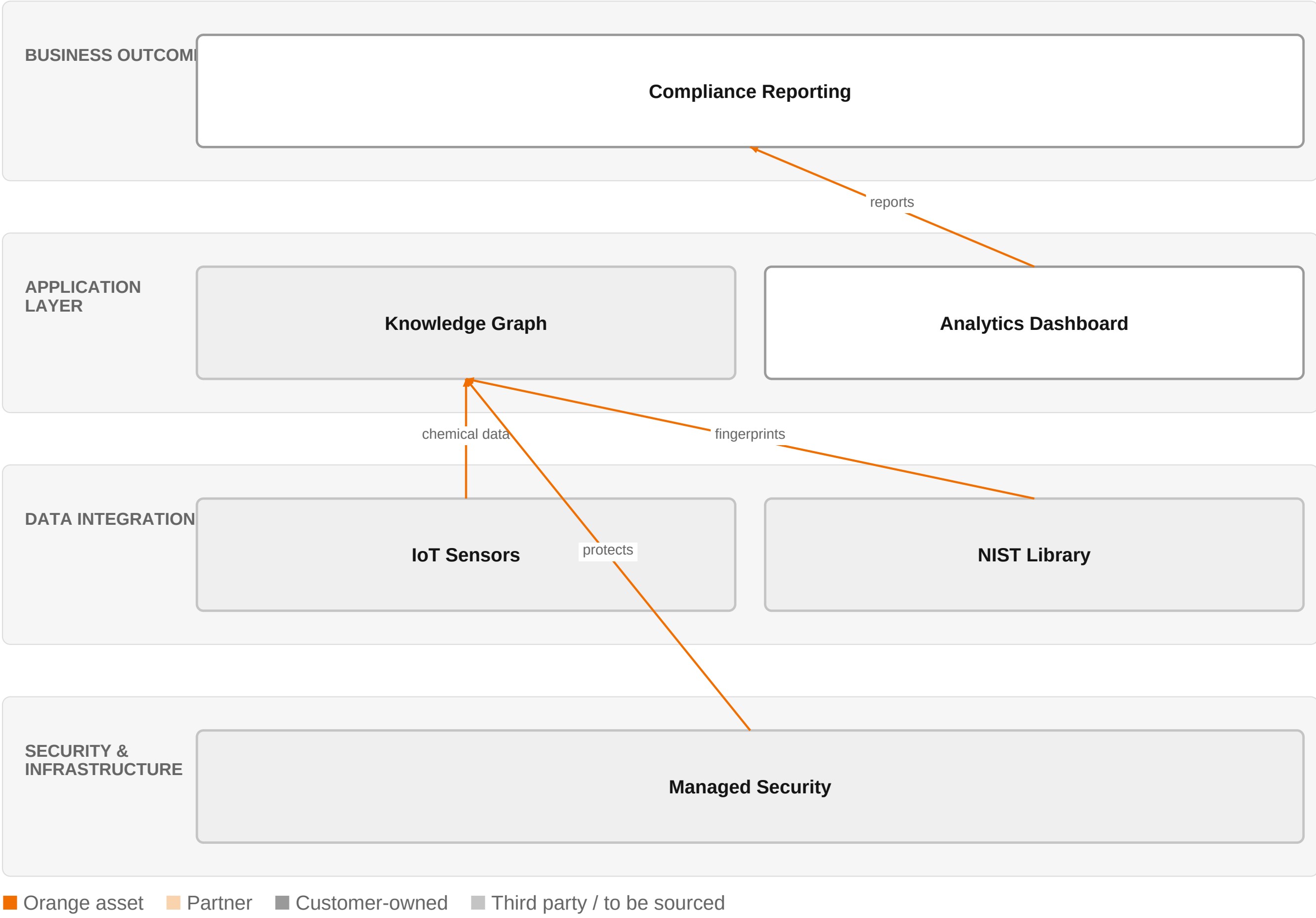
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

Chemical Compliance Knowledge Graph



This solution combines IoT data collection with NIST's chemical fingerprints in a knowledge graph, secured by Orange Cyberdefense, to streamline compliance reporting.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

Who buys, and why now

The primary buyer is the VP of Quality or Compliance in a manufacturing company, who feels the pain of manual, time-consuming substance identification and reporting. The trigger is a regulatory audit, a product recall, or a new regulation requiring disclosure of chemical substances. The operational pain includes slow lab analysis, high cost of external testing, and risk of non-compliance penalties.

Why it is hot now — every claim bound to a dated source

- NIST's library of chemical fingerprints helps manufacturers identify unknown compounds in food, drugs, cosmetics, and the environment. [SIG-7B84CE15615E]

What Orange would deliver, and why it can win

Why Orange can win

Orange brings IoT expertise and Orange Group researchers, which are relevant for building a knowledge graph and semantic modelling. Additionally, Orange Cyberdefense managed services provide a security layer that is critical for handling sensitive chemical data. However, the assets are thin: there is no specific chemical or compliance domain expertise, and the reference customers are not in this space. Orange would need to partner or build credibility.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed

ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 4.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
IBM	Global systems integrator	sells Knowledge graph and semantic modelling	structural	—
Palantir	Data and AI platform	sells Knowledge graph and semantic modelling; competes in OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—
Bosch	Industrial / OT specialist	active in Manufacturing; competes in OX: Smart Industries	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. How do you currently identify unknown substances in your products or supply chain?
2. What regulatory reporting obligations do you have for chemical substances, and how often do you need to report?
3. Have you experienced a compliance issue or recall due to an unidentified substance in the past year?
4. Do you have a centralized database of chemical substances used in your manufacturing processes?
5. How do you handle data privacy and security when sharing chemical data with external labs or partners?

Objections you will meet

They will say	You can answer
We already use an external lab for substance identification; why change?	That's a valid point. External labs are reliable, but they can be slow and costly. Our solution aims to complement your lab work by providing faster in-house screening using NIST's expanded library, reducing turnaround time and cost.

We're not sure the NIST library covers all the substances we deal with.	You're right that the library may not cover everything. However, it's expanding, and we can integrate it with your existing data and other sources. The knowledge graph approach allows you to add your own substances and link them to the library.
We're concerned about data security and privacy with IoT sensors and cloud processing.	That's a legitimate concern. We address it with Orange Cyberdefense managed services, which provide ISO 27001 certified security, and we can deploy on sovereign cloud if needed. Your data is protected end-to-end.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief that outlines how Orange Cyberdefense managed services, backed by ISO 27001 and TISAX certifications, can support a secure data environment for manufacturers using NIST's chemical fingerprint library, leveraging references from Saint-Gobain Glass, Skytale, and Everise to illustrate our expertise in secure data handling.
Sales	In a conversation with a quality or compliance leader at a consumer goods manufacturer, say: 'We've helped companies like Colgate-Palmolive and Mondelez International with their digital transformation, and we're exploring how to help manufacturers identify unknown substances for regulatory compliance using NIST's chemical fingerprint library—could we discuss your current challenges in this area?'
Strategist / Innovator	Commission a deep-dive brief on how Orange's knowledge graph and semantic modelling capabilities could be paired with NIST's chemical fingerprint library to create a compliance reporting service for manufacturers, using Orange Group researchers and IoT experts to assess technical feasibility.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-7B84CE15615E	2026-06-09	nist.gov	1	NIST Expands Its Library of 'Chemical Fingerprints' to Identify Unknown Substances
SIG-F633044AF87A	2026-01-29	openalex.org	1	SecuFL-IoT: an adaptive privacy-preserving federated learning framework for anomaly detection i...
SIG-E0AA4F59988C	2025-12-10	openalex.org	1	Anomaly detection for industrial applications: challenges, solutions, and future directions
SIG-3C2F7CAC7510	2025-10-01	openalex.org	1	Federated multi scale vision transformer with adaptive client aggregation for industrial defect...
SIG-A06EDB8B4416	2026-04-22	arxiv.org	3	IoT-Enhanced CNN-Based Labelled Crack Detection for Additive Manufacturing Image Annotation in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: summary (named entities that were never supplied: NIST); what_is_changing (named entities that were never supplied: NIST); what_orange_would_deliver (named entities that were never supplied: NIST); competitive_landscape (no claim survived evidence binding); risks_and_unknowns (named entities that were never supplied: NIST); diagram (3 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:25+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.